



Kadir Has University – Computer Engineering Department

CCIP503 Computer Networks & Security

Week – 1

Introduction,

Computer Networks and the Internet

**Assist. Prof. Dr.
İlktan AR**





0.1 Course - Introduction

Course Code	CCIP 503
Course Name	Computer Networks & Security
Local / ECTS Credit	3 / 7.5
Course Web Site	https://learn.khas.edu.tr Lecture notes, announcements, examples, etc.

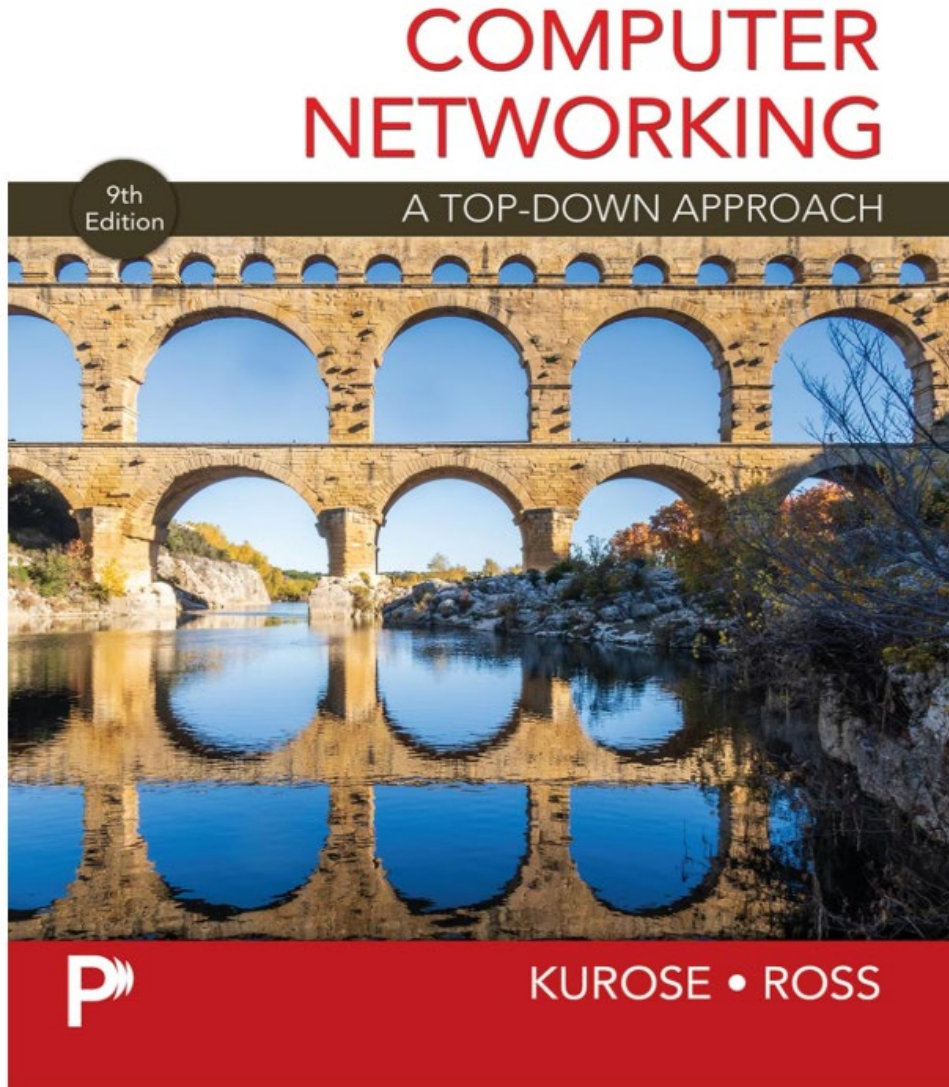
2026-2027 Fall Semester	
Instructor	Asst. Prof. Dr. İlktan Ar
E-mail	ilktana@khas.edu.tr
Phone	+90 212 533 65 32 (1206)
Office	D-252
Office Hours	Friday 12:00-13:00

0.2 Course - Schedule

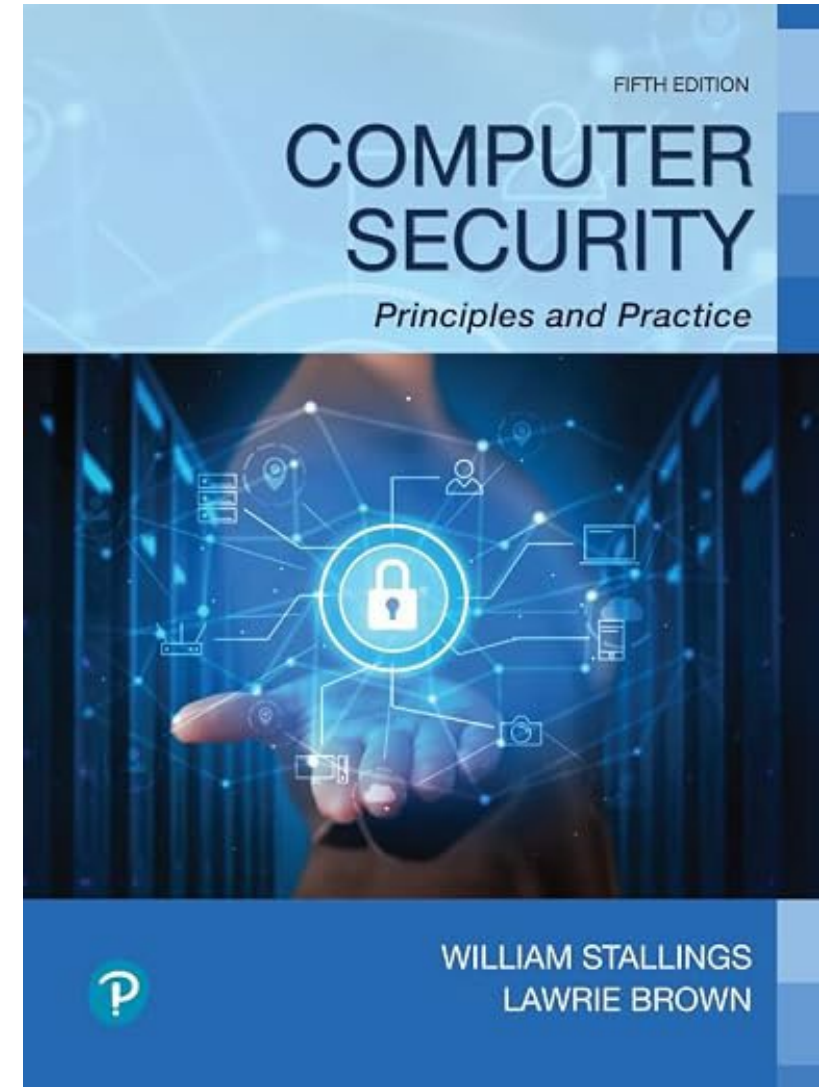
- 3 hours theory session in each week.
- Theory sessions: Mondays 15:00–18:00 @ PC-Lab 3.



0.3 Course - Textbooks



James F. Kurose, Keith W. Ross, "Computer Networking, A Top-Down Approach", Pearson Education Inc., 9th Ed., 2025.



William Stallings and Lawrie Brown, "Computer Security: Principles and Practice", Pearson Education Inc., 5th Ed., 2023.



0.4 Course - Textbooks

Additional Resources:

- Computer Networks, 5th Edition by Andrew S. Tanenbaum, David J. Wetherall, Pearson Education Inc.
- Communication Networks: A First Course, 2nd Edition by Jean Walrand, McGraw-Hill.
- Data Networks, 2nd Edition by Dimitri Bertsekas & Robert Gallager, Prentice Hall.



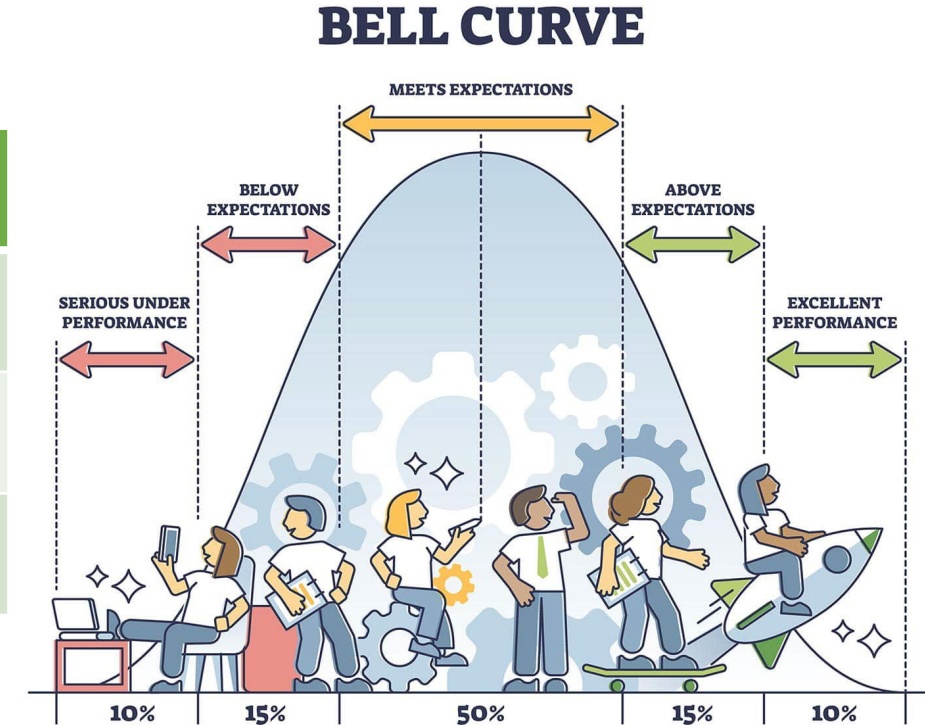
0.5 Course – Learning Outcomes

- Explain the fundamental concepts, protocols, and technologies underlying modern computer networks.
- Analyze the operation and performance of application, transport, network, and link layer protocols using appropriate concepts.
- Apply network analysis techniques by the Wireshark application for capturing, analyzing, and interpreting network traffic packets.
- Explain and apply fundamental computer security principles, technologies, and practices to identify common security threats and support the protection of computer networks and systems.



0.6 Course - Grading

Semester Studies	Percentage (%)
First Midterm Exam	20
Second Midterm Exam	30
Final Exam	50



- The conversion of the final grade (total score) into a letter grade will be carried out using Relative Grading System (Curve).



0.7 Course - Attendance

- Attendance will be collected but not graded.
- However, students are encouraged to attend all lectures.
- Attendance will enhance your level of understanding and keep you on track.



0.8 Course – Tentative Outline

Week	Date	Topic
1	28 Sept.	Computer Networks and the Internet
2	5 Oct.	Computer Networks and the Internet
3	12 Oct.	Application Layer
4	19 Oct.	Application Layer
5	26 Oct.	Transport Layer, Wireshark usage and examples
6	2 Nov.	First Midterm Exam
7	9 Nov.	Transport Layer
8	16 Nov.	The Network Layer: Data Plane
9	23 Nov.	The Network Layer: Control Plane
10	30 Nov.	The Link Layer and Local Area Networks
11	7 Dec.	Second Midterm Exam
12	14 Dec.	Security Technology and Principles
13	21 Dec.	Security Technology and Principles
14	28 Dec.	Security Technology and Principles, Talk for Final Exam
FN	???	Final Exam



Any Questions



Week 1 - Outline

01

What is the Internet?

02

The Network Edge

03

The Network Core

04

Next Week



1.0 Computer Network

- A **network** is a set of devices (often referred to as nodes) connected by communication links.
- A **node** can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.
- Classification of Computer Networks:
 - Based on Transmission Mode: Simplex, Half-Duplex, Full-Duplex
 - Based on Time in Transmission Type: Synchronous Transmission, Asynchronous Transmission
 - Based on Authentication: Peer to Peer Connection, Server Based Connection
 - Based on Geographical location: Local Area Networks (LAN), Wide Area Network (WAN), MAN (Metropolitan Area Network)...
 - Based on Reliability: Connection-oriented, Connectionless

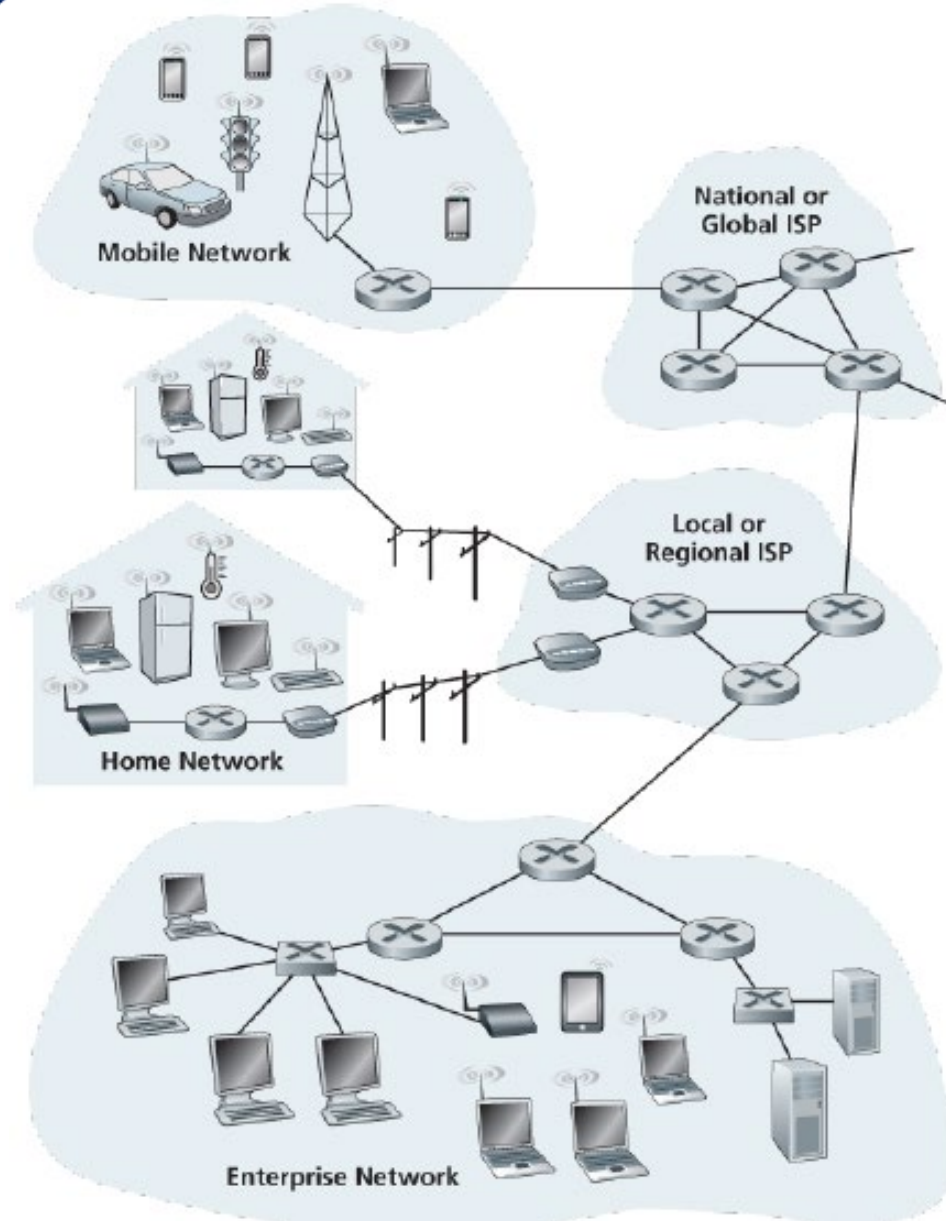
1.1 The Internet

- The Internet is a computer network that interconnects billions of computing devices throughout the world.
- The Internet, sometimes called simply "the Net," is a worldwide system of computer networks - a network of networks.
- The Internet is the infrastructure, and the Web is a service on top.

Worldwide Internet users

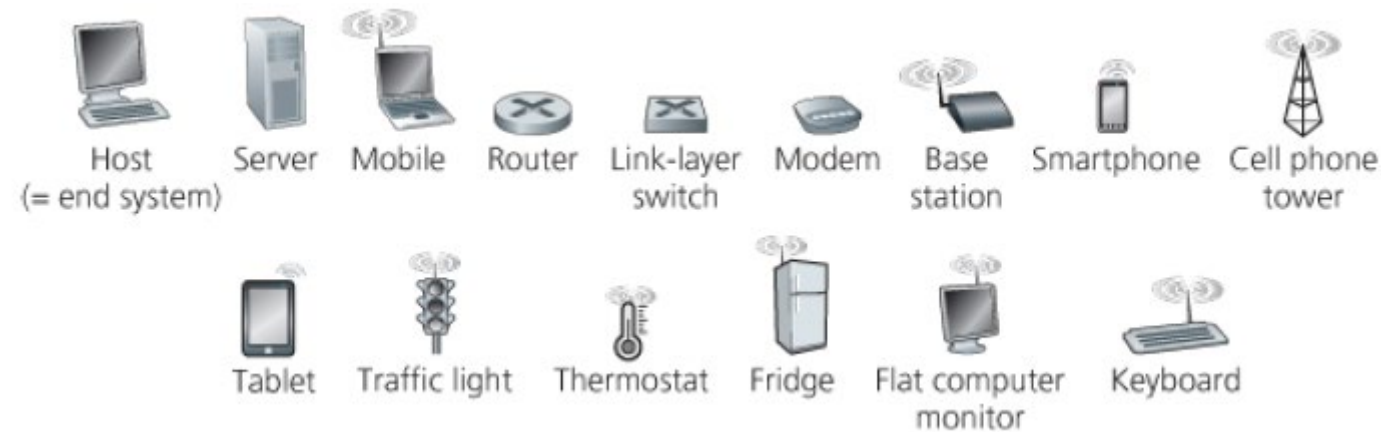
Users	2005	2010	2017	2019	2021
World population	6.5 billion	6.9 billion	7.4 billion	7.75 billion	7.9 billion
Worldwide	16%	30%	48%	53.6%	63%
In developing world	8%	21%	41.3%	47%	57%
In developed world	51%	67%	81%	86.6%	90%

1.2 Network of Networks...



A **host** is any hardware device that has the capability of permitting access to a network via a user interface, specialized software, network address, protocol stack, or any other means. Briefly **host** is any computing device connected to a network...

hosts = end systems.



1.3 Internet-Connected Devices



IP picture frame
<http://www.ceiva.com/>



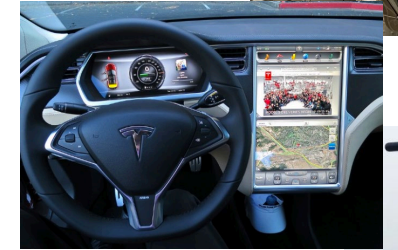
Web-enabled toaster +
weather forecaster



Tweet-a-watt:
monitor energy use



bikes



cars



scooters



Slingbox: watch,
control cable TV remotely



sensorized,
bed
mattress



Internet phones



Gaming devices

Internet
refrigerator



1.4 Some Terms

- End systems are connected together by a network of **communication links** and **packet switches**.
- There are many types of communication links, which are made up of different types of physical media, including coaxial cable, copper wire, optical fiber, and radio spectrum.
- Different links can transmit data at different rates, with the **transmission rate** of a link measured in bits/second.
- When one end system has data to send to another end system, the sending end system segments the data and adds header bytes to each segment. The resulting packages of information, known as **packets** in the jargon of computer networks, are then sent through the network to the destination end system, where they are reassembled into the original data.



1.5 Packet-Switched Networks

- Packet-switched networks (which transport packets) are in many ways like transportation networks of highways, roads, and intersections (which transport vehicles).
- Consider, for example, a factory that needs to move a large amount of cargo to some destination warehouse located thousands of kilometers away.
- At the factory, the cargo is segmented and loaded into a fleet of trucks. Each of the trucks then independently travels through the network of highways, roads, and intersections to the destination warehouse.
- At the destination warehouse, the cargo is unloaded and grouped with the rest of the cargo arriving from the same shipment.
- **Packets** are analogous to trucks, **communication links** are analogous to highways and roads, **packet switches** are analogous to intersections, and **end systems** are analogous to buildings. Just as a truck takes a path through the transportation network, a packet takes a **path** through a **computer network**.

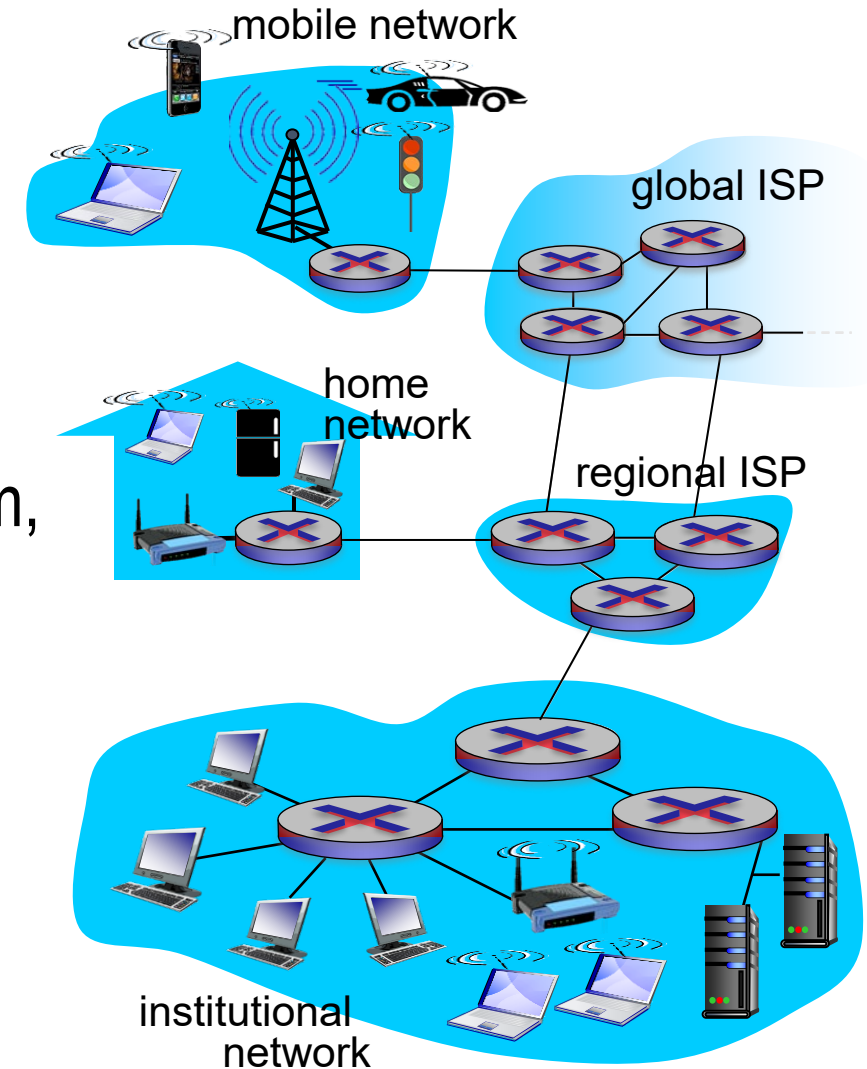


1.6 Some Terms (conts.)

- End systems access the Internet through **Internet Service Providers (ISPs)**, including residential ISPs such as local cable or telephone companies; corporate ISPs; university ISPs; ISPs that provide WiFi access in airports, hotels, coffee shops, and other public places; and cellular data ISPs, providing mobile access to our smartphones and other devices.
- End systems, packet switches, and other pieces of the Internet run **protocols** that control the sending and receiving of information within the Internet. The **Transmission Control Protocol (TCP)** and the **Internet Protocol (IP)** are two of the most important protocols on the Internet.
- **Internet standards** are developed by the **Internet Engineering Task Force (IETF)**. The IETF standards documents are called **requests for comments (RFCs)**.

1.7 Briefly...

- *Internet*: “network of networks”
 - Interconnected ISPs
- *protocols* control sending, receiving of messages
 - e.g., TCP, IP, HTTP (Web), streaming video, Zoom, Skype, 802.11
- *Internet standards*
 - RFC: Request for comments
 - IETF: Internet Engineering Task Force

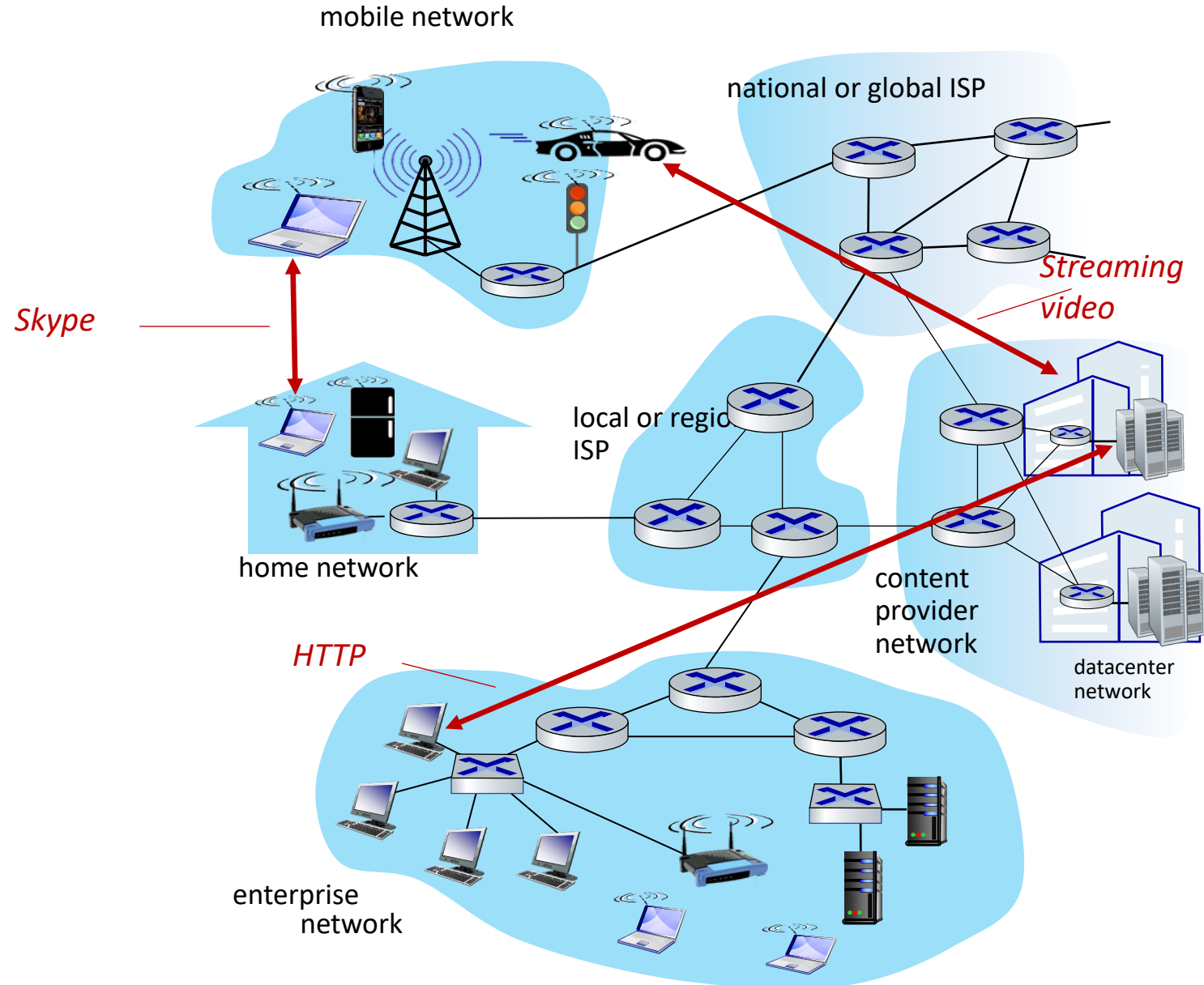




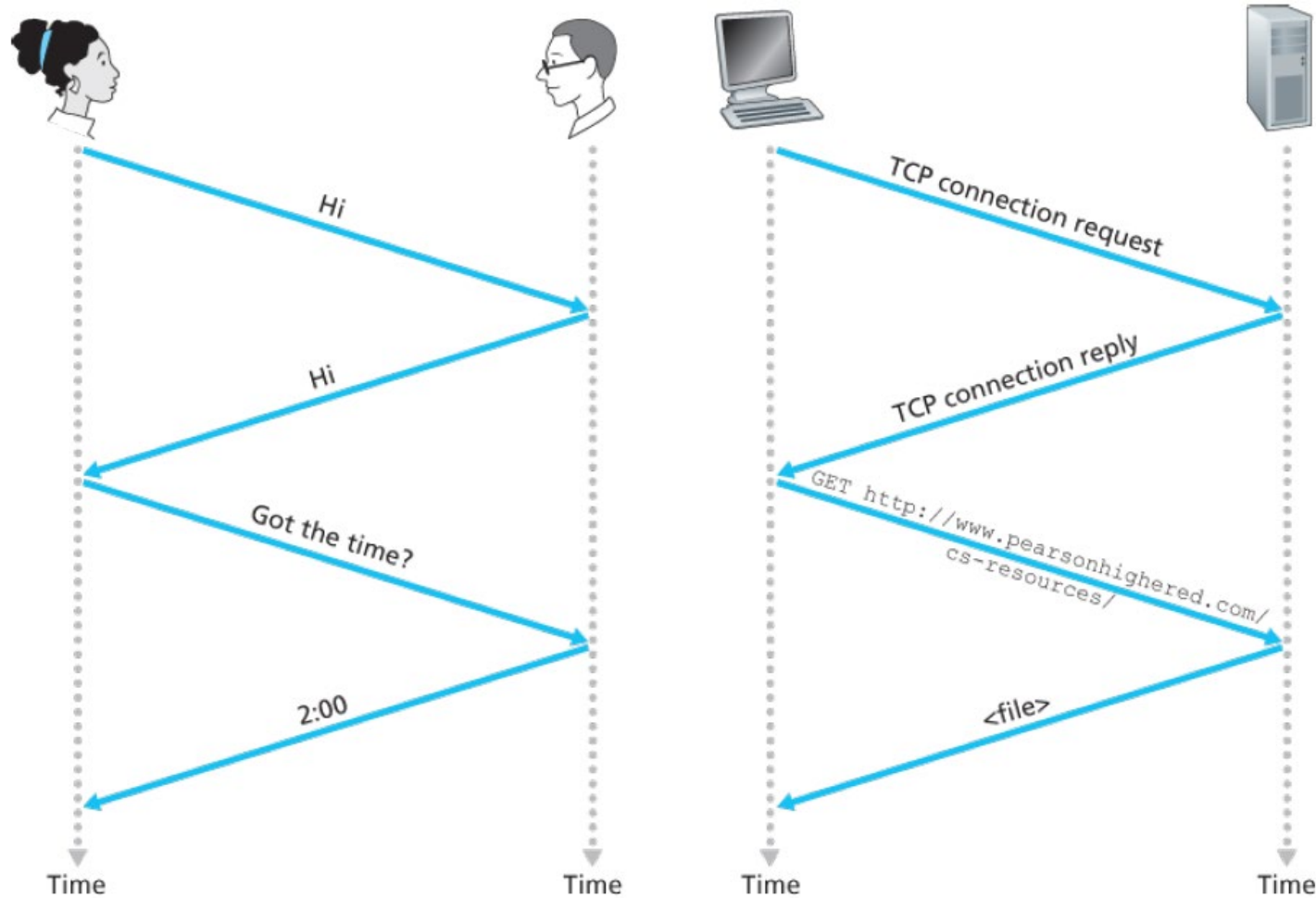
1.8 Services

- We can also describe the Internet from an entirely different angle—namely, as *an infrastructure that provides services to applications*.
- In addition to traditional applications such as e-mail and Web surfing, Internet applications include mobile smartphone and tablet applications, including Internet messaging, mapping with real-time road-traffic information, music streaming from the cloud, movie and television streaming, online social networks, video conferencing, multi-person games, and location-based recommendation systems.
- The applications are said to be **distributed applications**, since they involve multiple end systems that exchange data with each other.
- End systems attached to the Internet provide a **socket interface** that specifies how a program running on one end system asks the Internet infrastructure to deliver data to a specific destination program running on another end system.

1.9 The Internet: a “services” view



1.10 What Is a Protocol?



A human protocol and a computer network protocol

- A **network protocol** is similar to a human protocol, except that the entities exchanging messages and taking actions are hardware or software components of some device.



1.11 What Is a Protocol? (cont.)

Network protocols:

- computers (devices) rather than humans
- all communication activity in Internet governed by protocols

*Protocols define the **format, order** of **messages sent and received** among network entities, and **actions taken** on message transmission, receipt*

A vibrant, abstract graphic consisting of numerous overlapping paint splashes in red, yellow, blue, and green. The splashes are dynamic and energetic, creating a colorful burst effect.

Any Questions



TIME
FOR A
BREAK!



Week 1 - Outline

01

What is the Internet?

02

The Network Edge

03

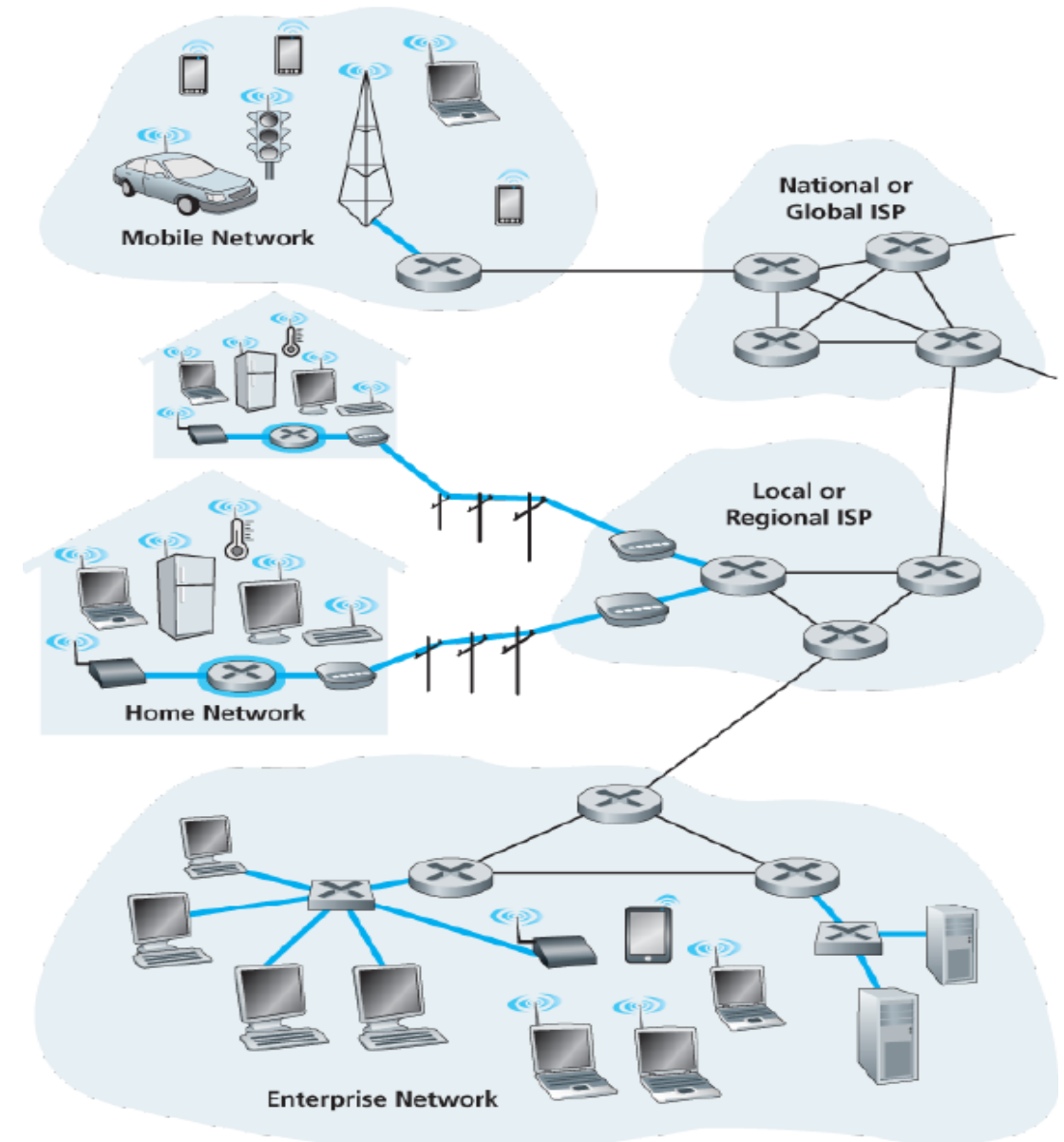
The Network Core

04

Next Week

2.1 A Closer Look at Network Structure

- host = end system
- Hosts are sometimes further divided into two categories: **clients** and **servers**.
- The **access network**—the network that physically connects an end system to the first router (also known as the “edge router”) on a path from the end system to any other distant end system. (blue shaded lines in figure)
- **Physical media**: wired, wireless communications links.
- **Network core**: Interconnected routers, network of networks.



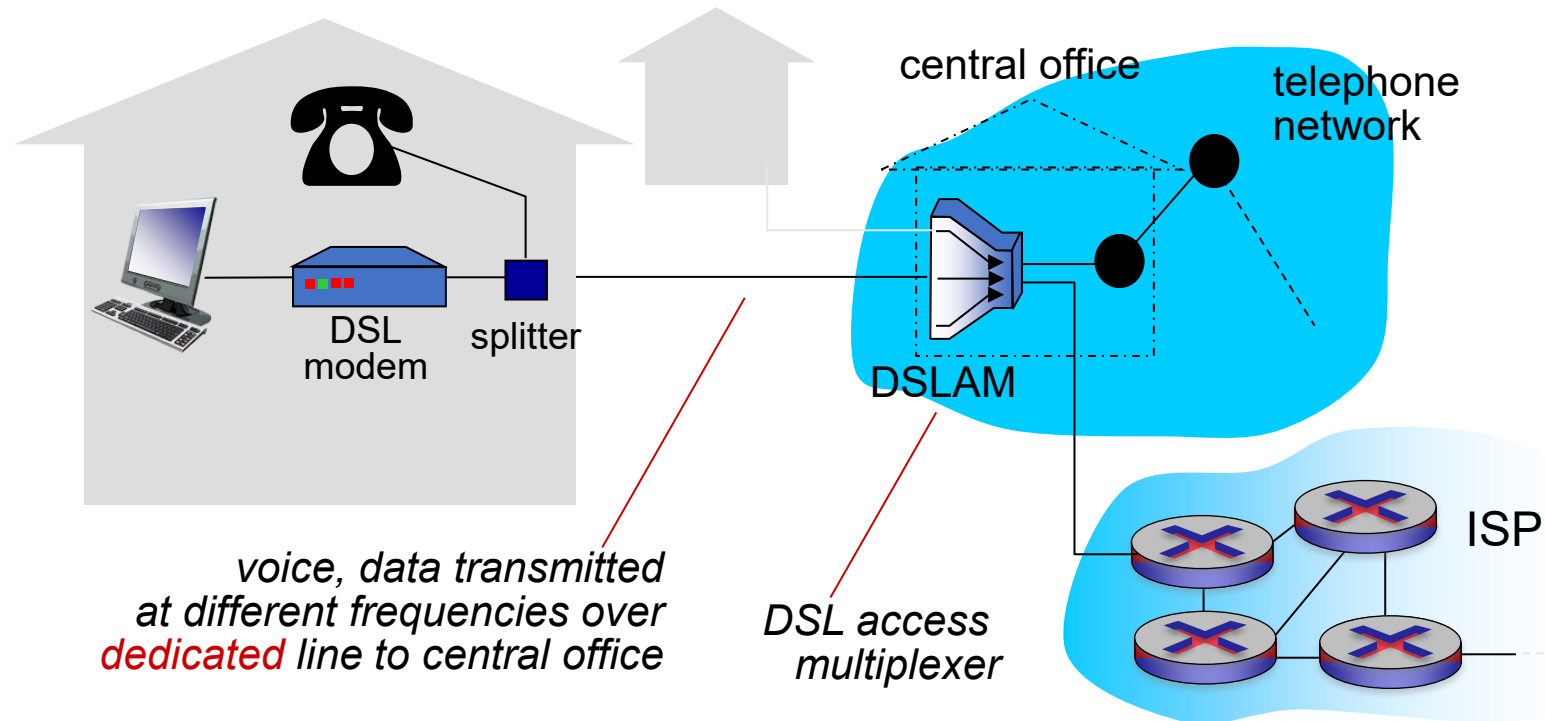


2.2 Access Networks

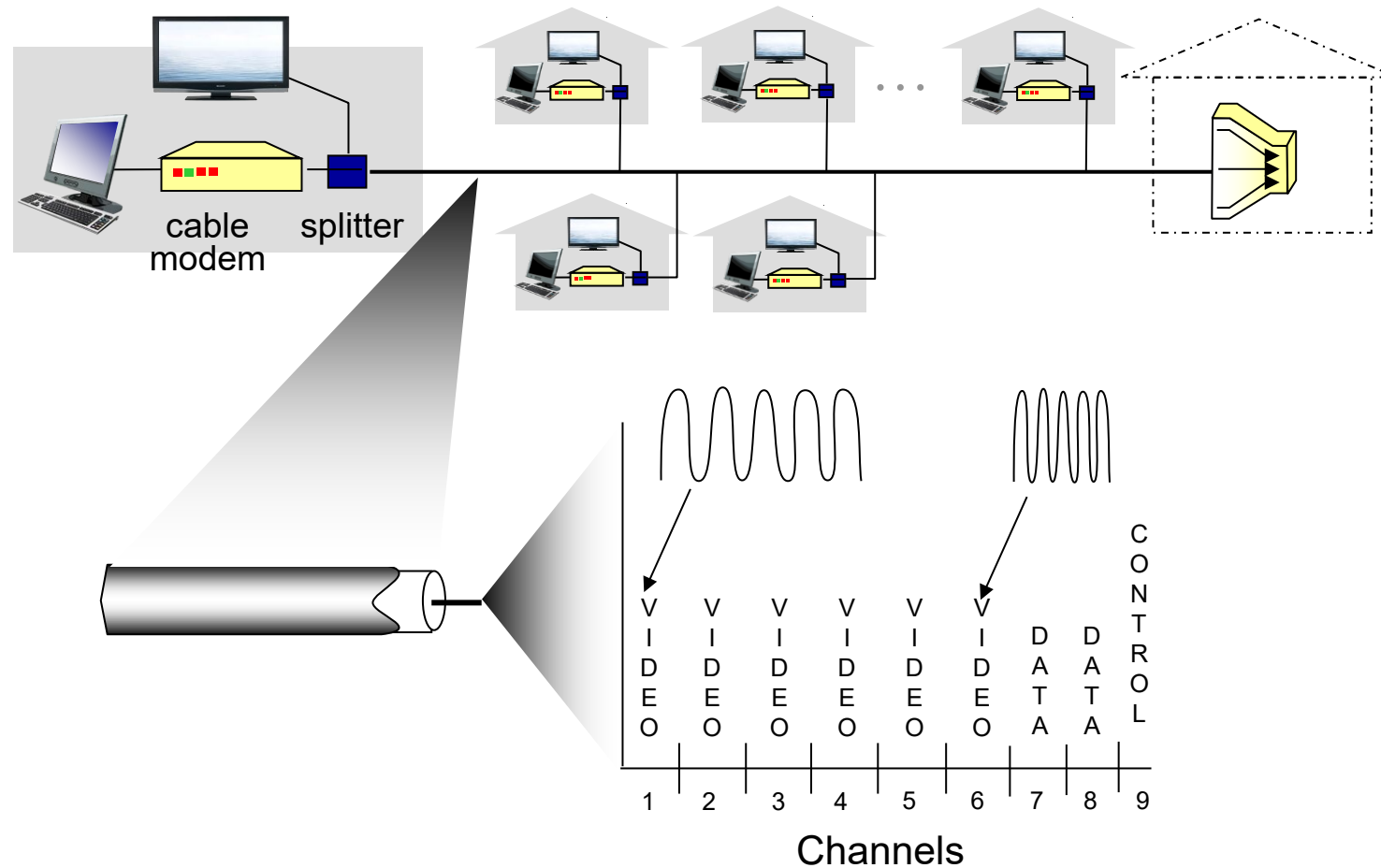
- Today, the two most popular types of broadband residential access are **digital subscriber line (DSL)** and cable.
- The residential telephone line carries both data and traditional telephone signals simultaneously, which are encoded at different frequencies:
 - A high-speed downstream channel, in the 50 kHz to 1 MHz band
 - A medium-speed upstream channel, in the 4 kHz to 50 kHz band
 - An ordinary two-way telephone channel, in the 0 to 4 kHz band

2.3 Access Network: Digital Subscriber Line (DSL)

- Use **existing** telephone line to central office DSLAM (Digital Subscriber Line Access Multiplexer)
 - data over DSL phone line goes to Internet
 - voice over DSL phone line goes to telephone net
- 24-52 Mbps dedicated downstream transmission rate
- 3.5-16 Mbps dedicated upstream transmission rate



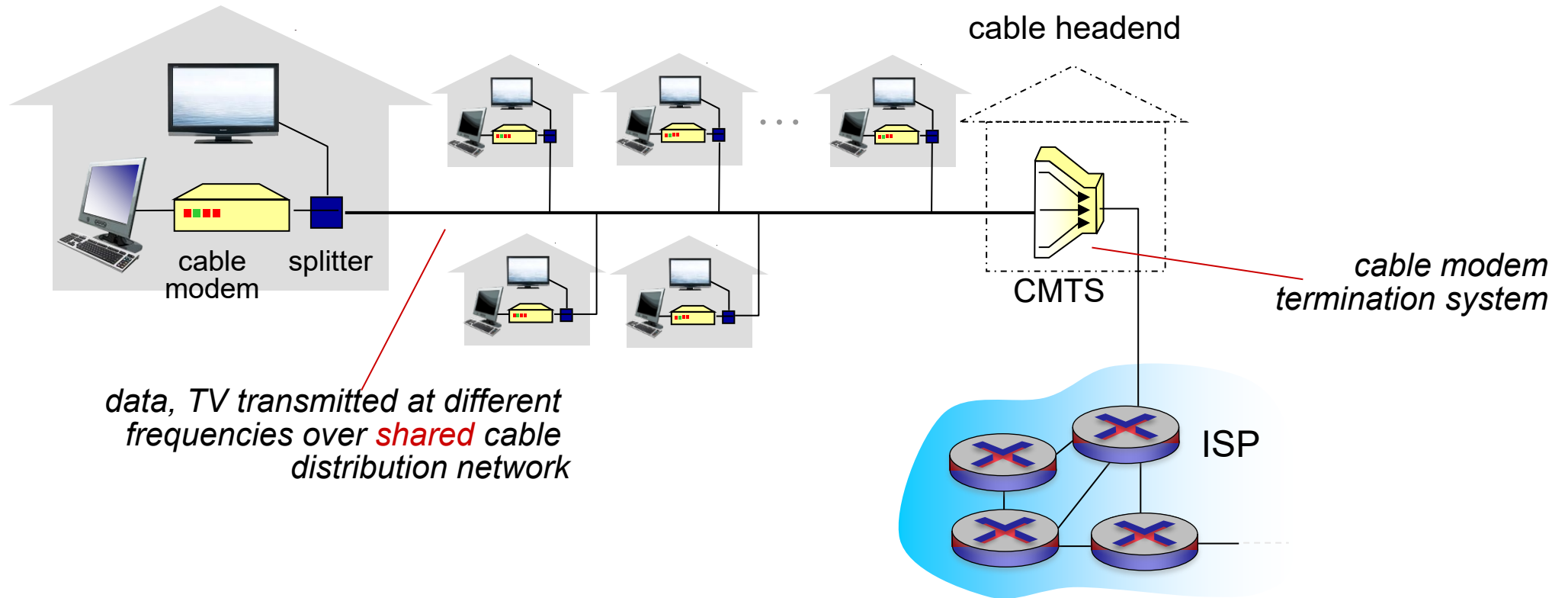
2.4 Access Network: Cable Network



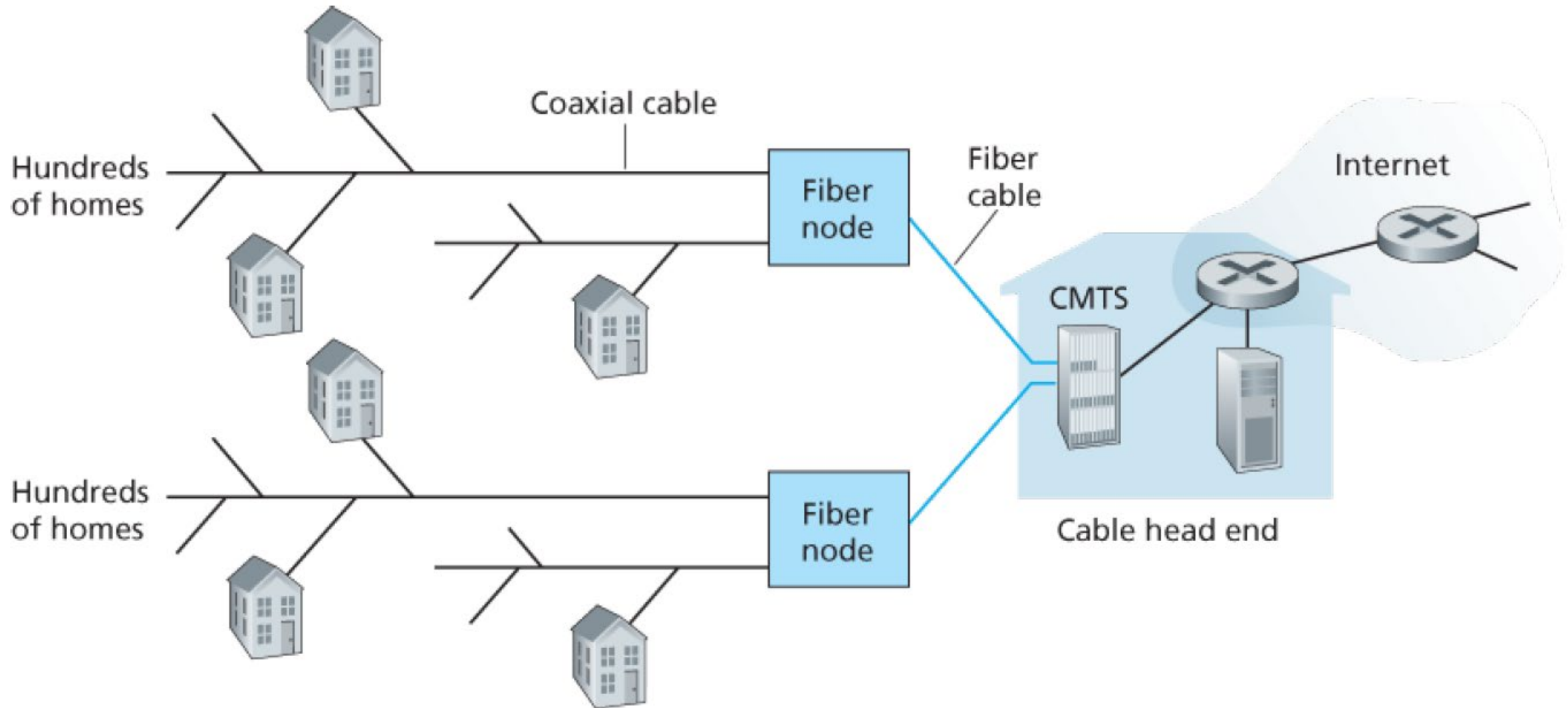
Frequency division multiplexing: different channels transmitted in different frequency bands.

2.5 Access Network: Cable Network (cont.)

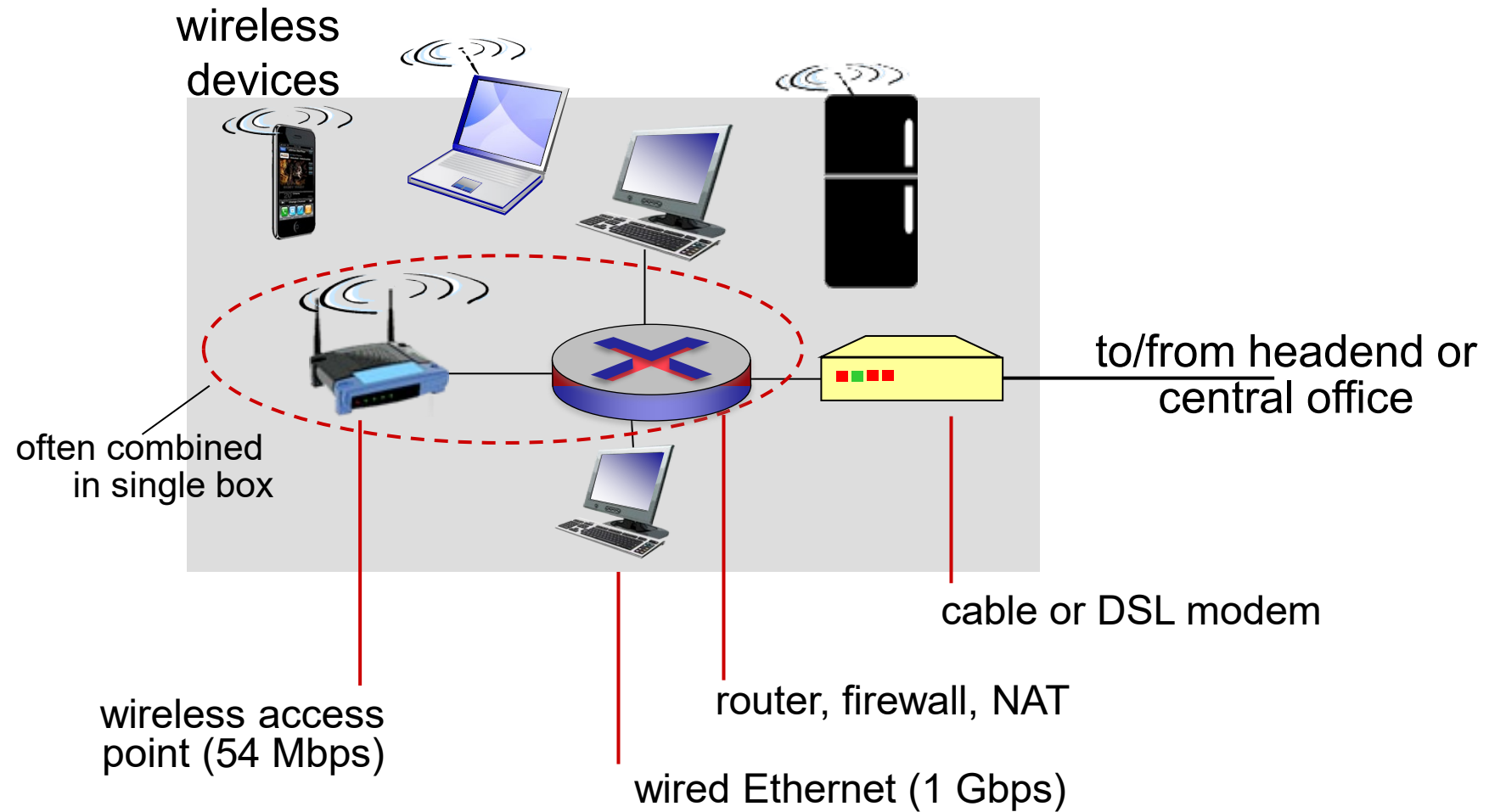
- Network of cable, fiber attaches homes to ISP router:
 - homes **share access network** to cable headend
 - unlike DSL, which has dedicated access to central office



2.6 A Hybrid Fiber-Coaxial Access Network

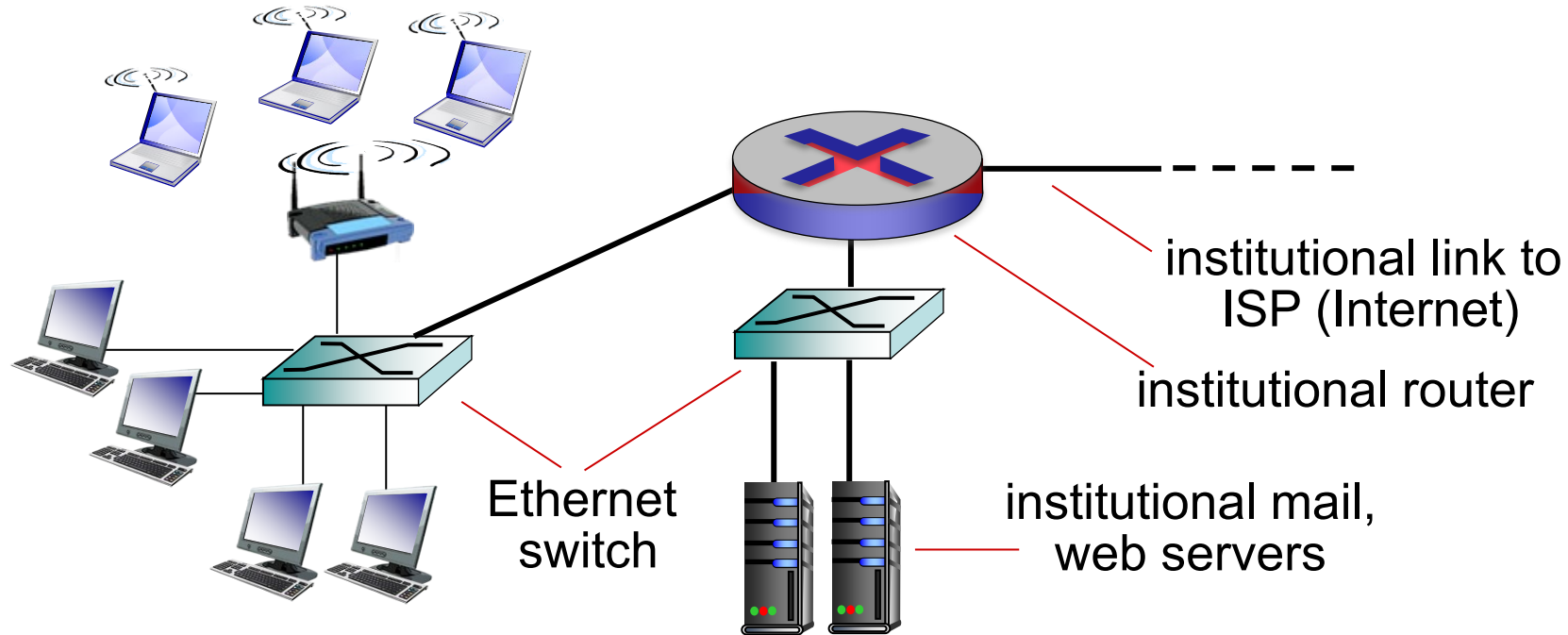


2.7 Access Network: Home Network



2.8 Enterprise Access Networks (Example)

- Typically used in companies, universities, etc.
- 10 Mbps, 100Mbps, 1Gbps, 10Gbps transmission rates.
- Today, end systems typically connect into Ethernet switch.



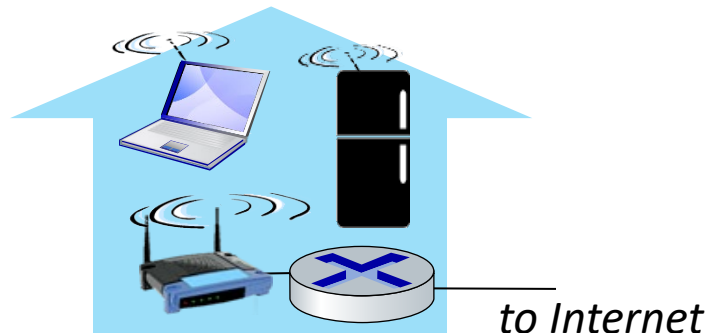
2.9 Wireless Access Networks

Shared *wireless* access network connects end system to router

- via base station aka “access point”

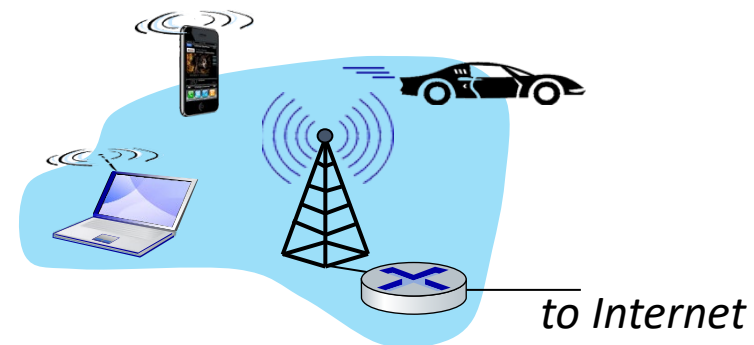
Wireless local area networks (WLANs)

- typically within or around building (~100 ft)
- 802.11b/g/n (WiFi): 11, 54, 450 Mbps transmission rate



Wide-area cellular access networks

- provided by mobile, cellular network operator (10's km)
- 10's Mbps
- 4G/5G cellular networks

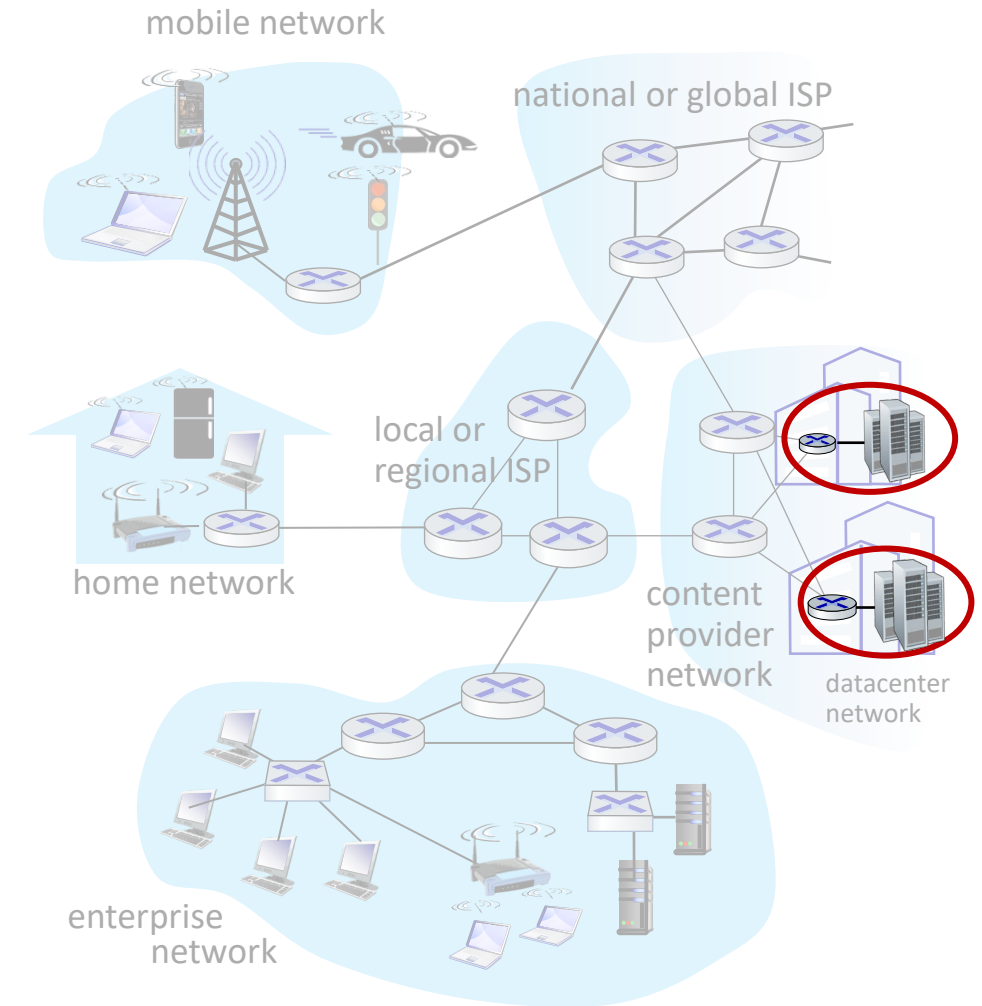


2.10 Access Networks: Data Center Networks

- high-bandwidth links (10s to 100s Gbps) connect hundreds to thousands of servers together, and to Internet



Courtesy: Massachusetts Green High Performance Computing Center (mghpcc.org)



2.11 Physical Media

- HFC (Hybrid Fiber-Coaxial) uses a combination of fiber cable and coaxial cable.
- DSL and Ethernet use copper wire.
- Mobile access networks use the radio spectrum.
- Data, when traveling from source to destination, passes through a series of transmitter-receiver pairs.
- For each transmitter receiver pair, the data is sent by propagating electromagnetic waves or optical pulses across a **physical medium**. (**Media** is the **plural** of **medium** !)
- The physical medium can take many shapes and forms and does not have to be of the same type for each transmitter-receiver pair along the path.
- Physical media fall into two categories: **guided media** and **unguided media**:
 - Guided media: the waves are guided along a solid medium, such as a fiber-optic cable, a twisted-pair copper wire, or a coaxial cable.
 - Unguided media: the waves propagate in the atmosphere and in outer space, such as in a wireless LAN or a digital satellite channel.

2.12 Physical Media: Twisted Pair, Coaxial, Fiber Optic

twisted pair (TP)

- two insulated copper wires
 - Category 5: 100 Mbps, 1 Gbps Ethernet
 - Category 6: 10Gbps



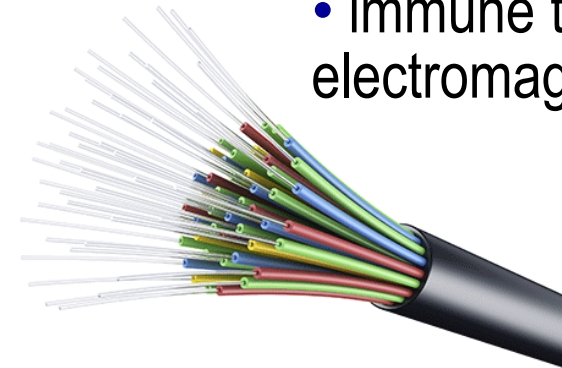
coaxial cable:

- two concentric copper conductors
- bidirectional
- broadband:
 - multiple channels on cable
 - HFC



fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
 - high-speed point-to-point transmission (e.g., 10's-100's Gbps transmission rate)
- low error rate:
 - repeaters spaced far apart
 - immune to electromagnetic noise



2.13 Physical Media: Radio

- signal carried in electromagnetic spectrum
- no physical “wire”
- Bidirectional (tr:çift yönlü)
- propagation environment effects:
 - reflection
 - obstruction by objects
 - interference

radio link types:

- terrestrial (tr:karasal) microwave
 - e.g. up to 45 Mbps channels
- LAN (e.g., WiFi)
54 Mbps
- wide-area (e.g., cellular)
 - 4G cellular: ~ 10 Mbps
- satellite
 - Kbps to 45Mbps channel (or multiple smaller channels)
 - 270 msec end-end delay
 - geosynchronous versus low altitude



Week 1 - Outline

02

The Network Edge

03

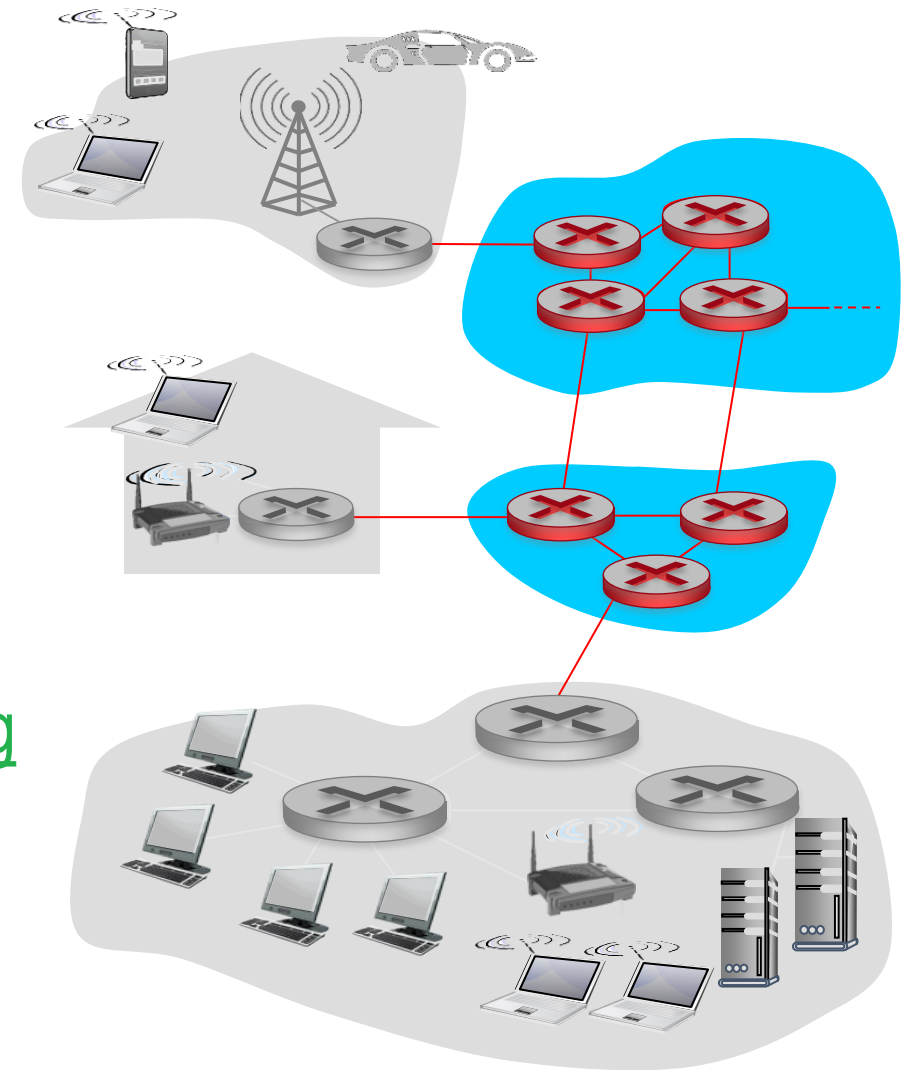
The Network Core

04

Next Week

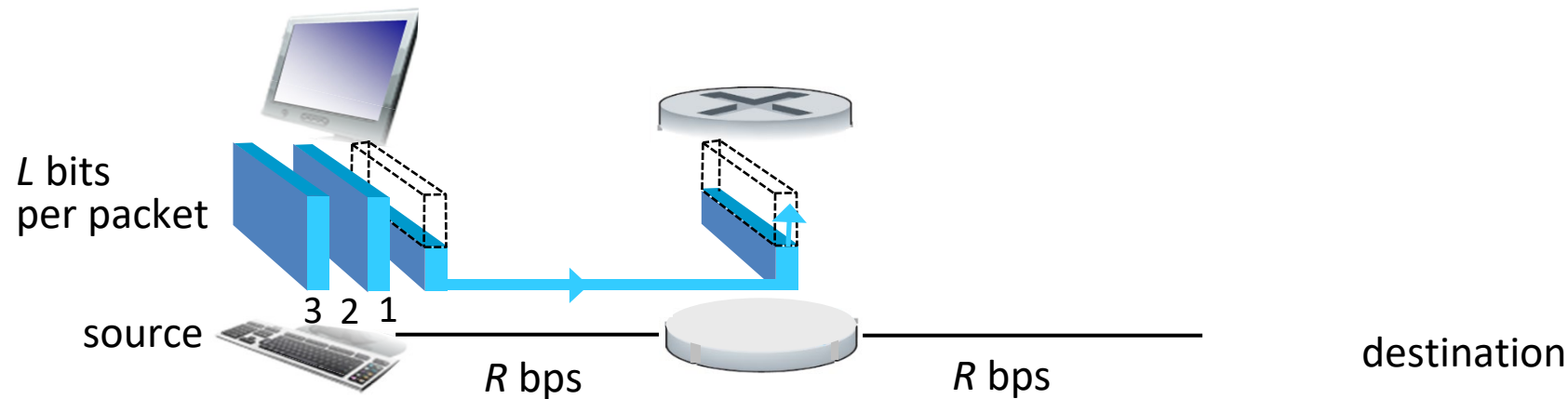
3.1 The Network Core

- Mesh of interconnected routers
- **Packet-switching: hosts break application-layer messages into *packets***
 - Forward packets from one router to the next, across links on path from source to destination
 - Each packet transmitted at full link capacity
- Packets are transmitted over each communication link at a rate equal to the *full* transmission rate of the link.
- So, if a source end system or a packet switch is sending a packet of L bits over a link with transmission rate R bits/sec, then the time to transmit the packet is L / R seconds.



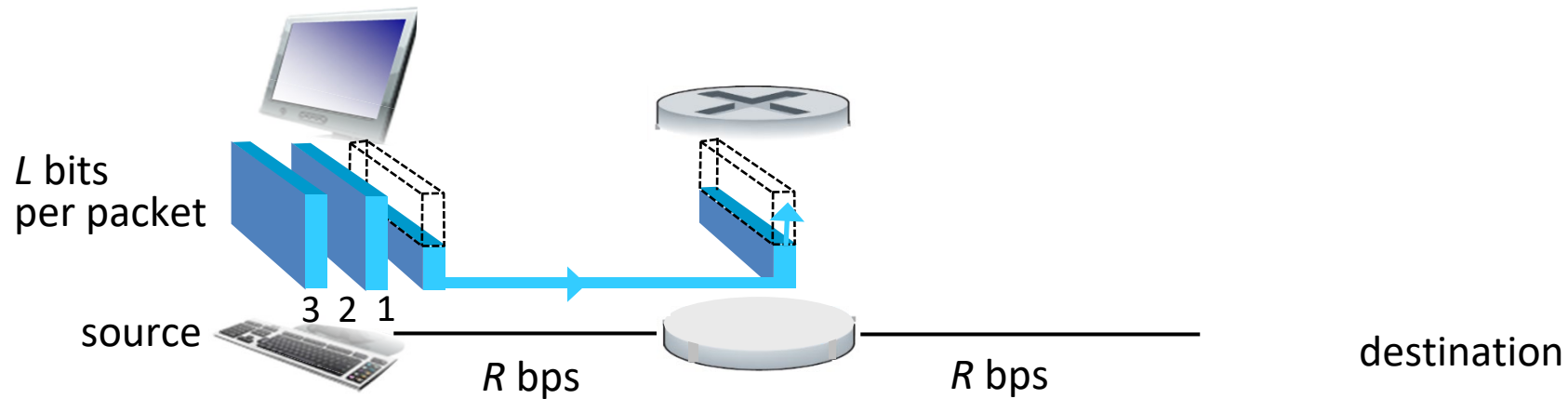
3.2 Packet-Switching: Store-and-Forward Transmission

- Takes L/R seconds to transmit (push out) L -bit packet into link at R bps.
- *Store and forward*: entire packet must arrive at router before it can be transmitted on next link.
- End-end delay = $2L/R$ (assuming zero propagation delay)



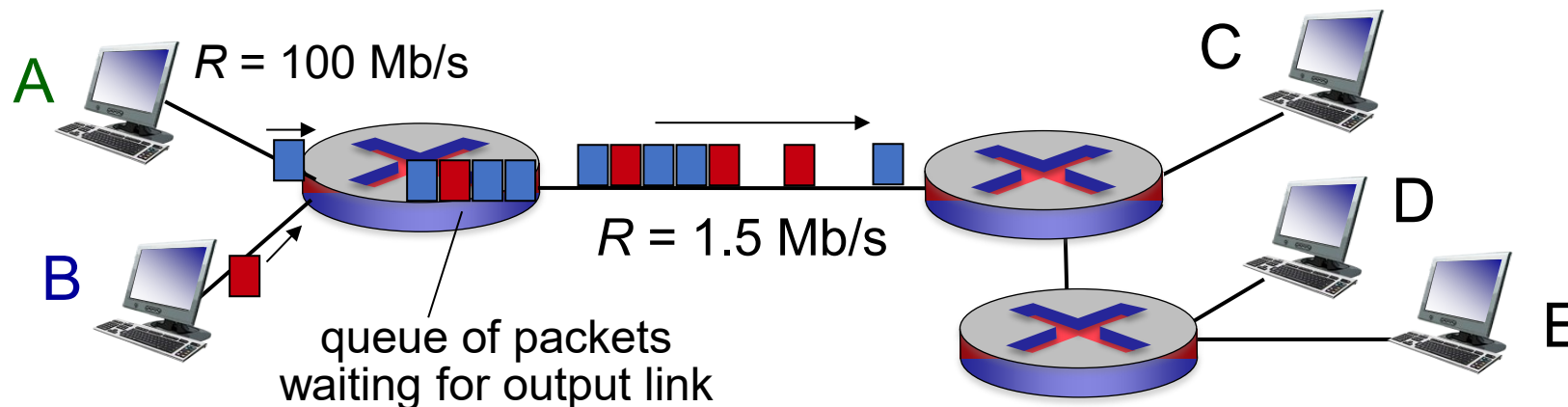
3.3 Example Question

- $L = 7.5$ Mbits
- $R = 1.5$ Mbps
- One-hop transmission delay = 5 sec (assuming zero propagation delay)
- Two--hop transmission delay = 10 sec (assuming zero propagation delay)

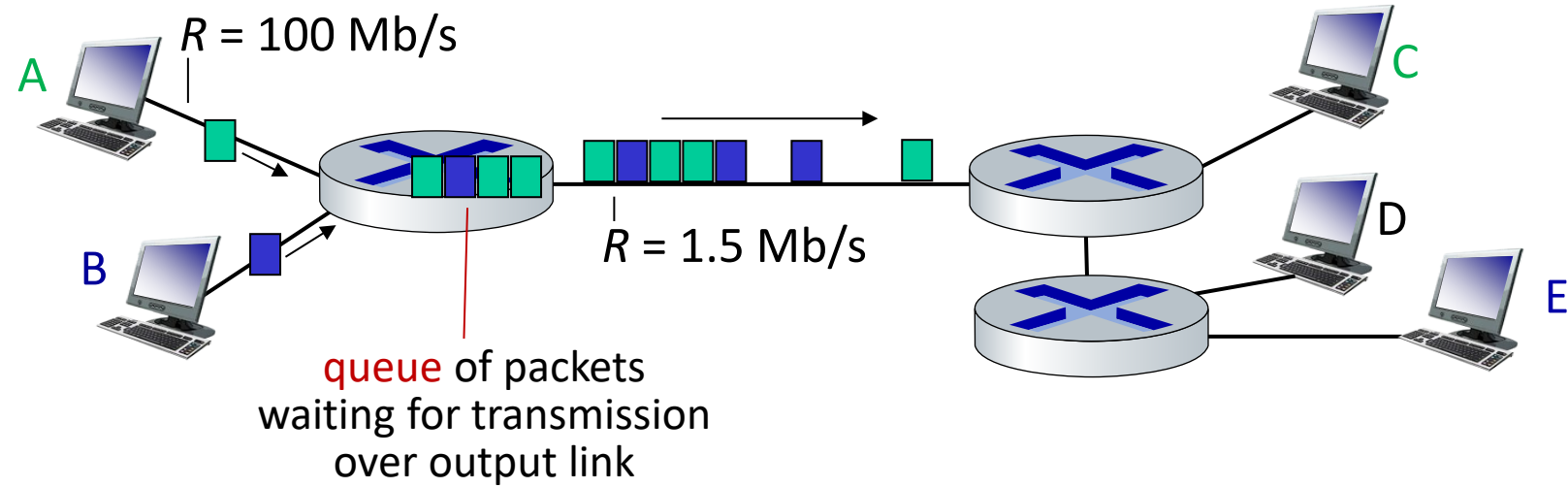


3.4 Packet Switching: Queueing Delay, Loss

- Each packet switch has multiple links attached to it. For each attached link, the packet switch has an **output buffer** (also called an **output queue**), which stores packets that the router is about to send into that link.
- The output buffers play a key role in packet switching. If an arriving packet needs to be transmitted onto a link but finds the link busy with the transmission of another packet, the arriving packet must wait in the output buffer.
- Thus, in addition to the store-and-forward delays, packets suffer output buffer **queueing delays**.
- Since the amount of buffer space is finite, an arriving packet may find that the buffer is completely full with other packets waiting for transmission. In this case, **packet loss** will occur—either the arriving packet or one of the already-queued packets will be dropped.



3.5 Queueing



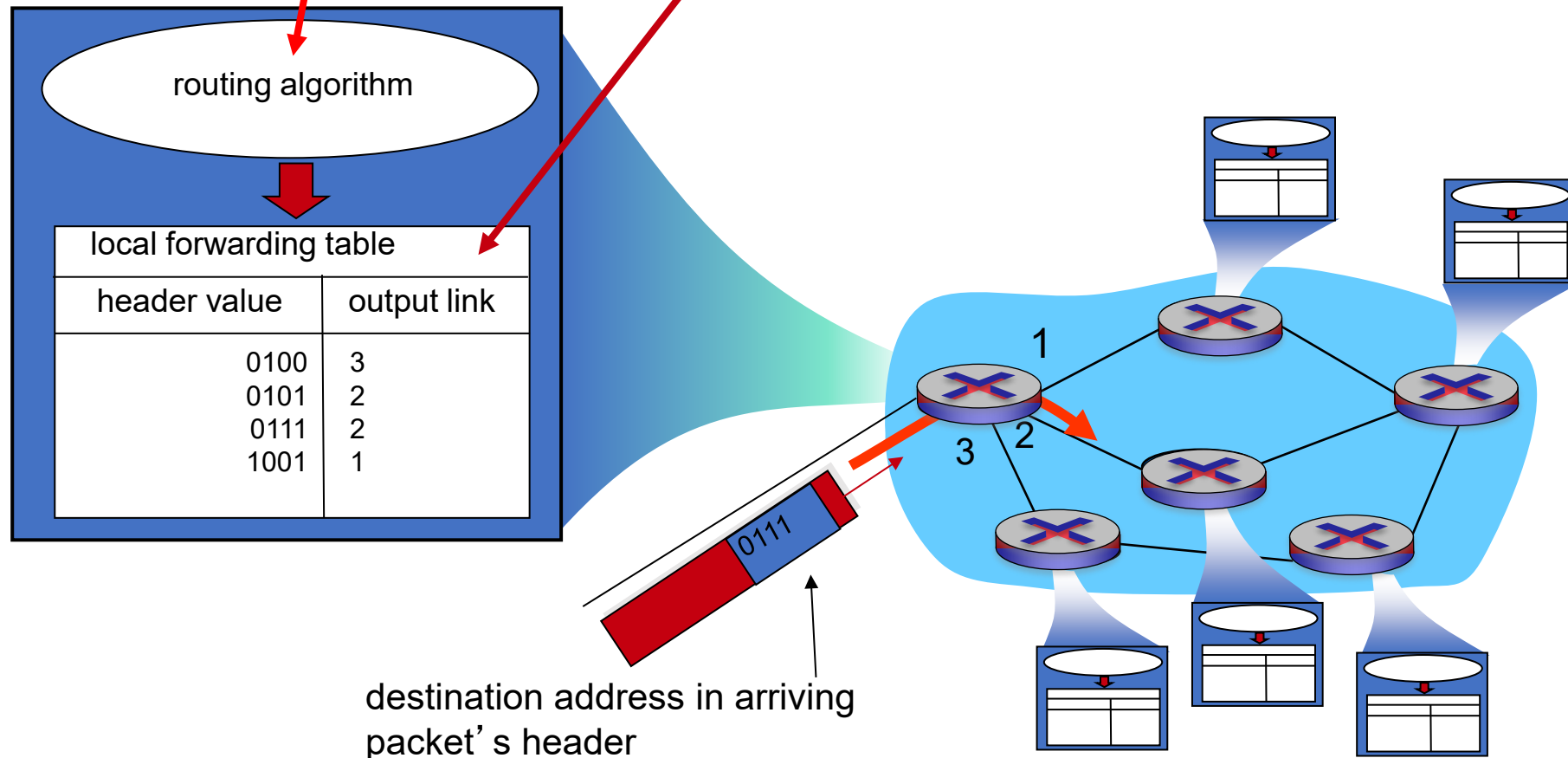
Queueing occurs when work arrives faster than it can be serviced:



3.6 Two Key Network-Core Functions

- **Routing**: determines source-destination route taken by packets
 - Routing Algorithms

Forwarding: move packets from router's input to appropriate router output



3.7 Two Key Network-Core Func: Routing



3.8 Two Key Network-Core Func: Forwarding

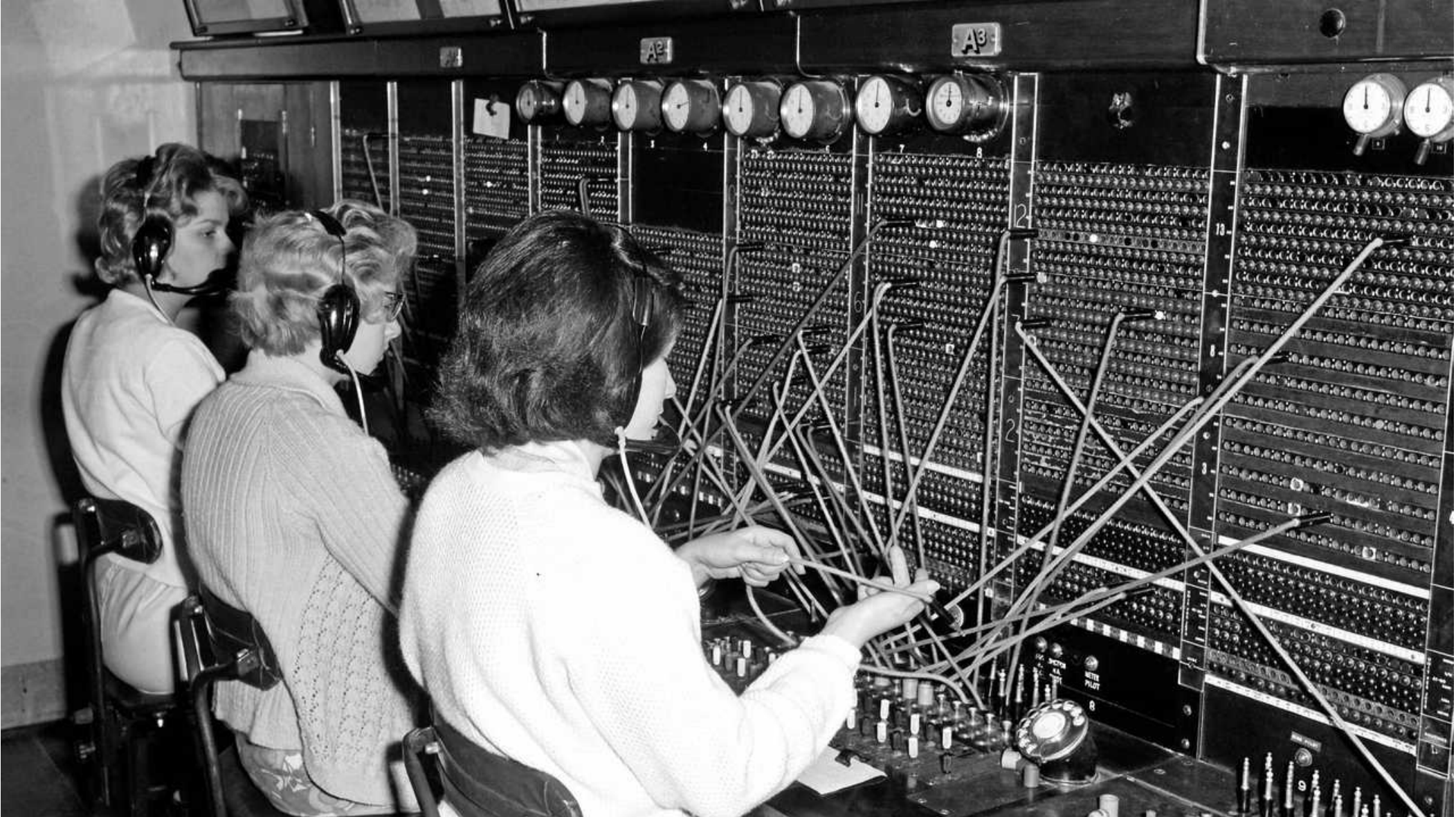




3.9 Alternative Core: Circuit Switching

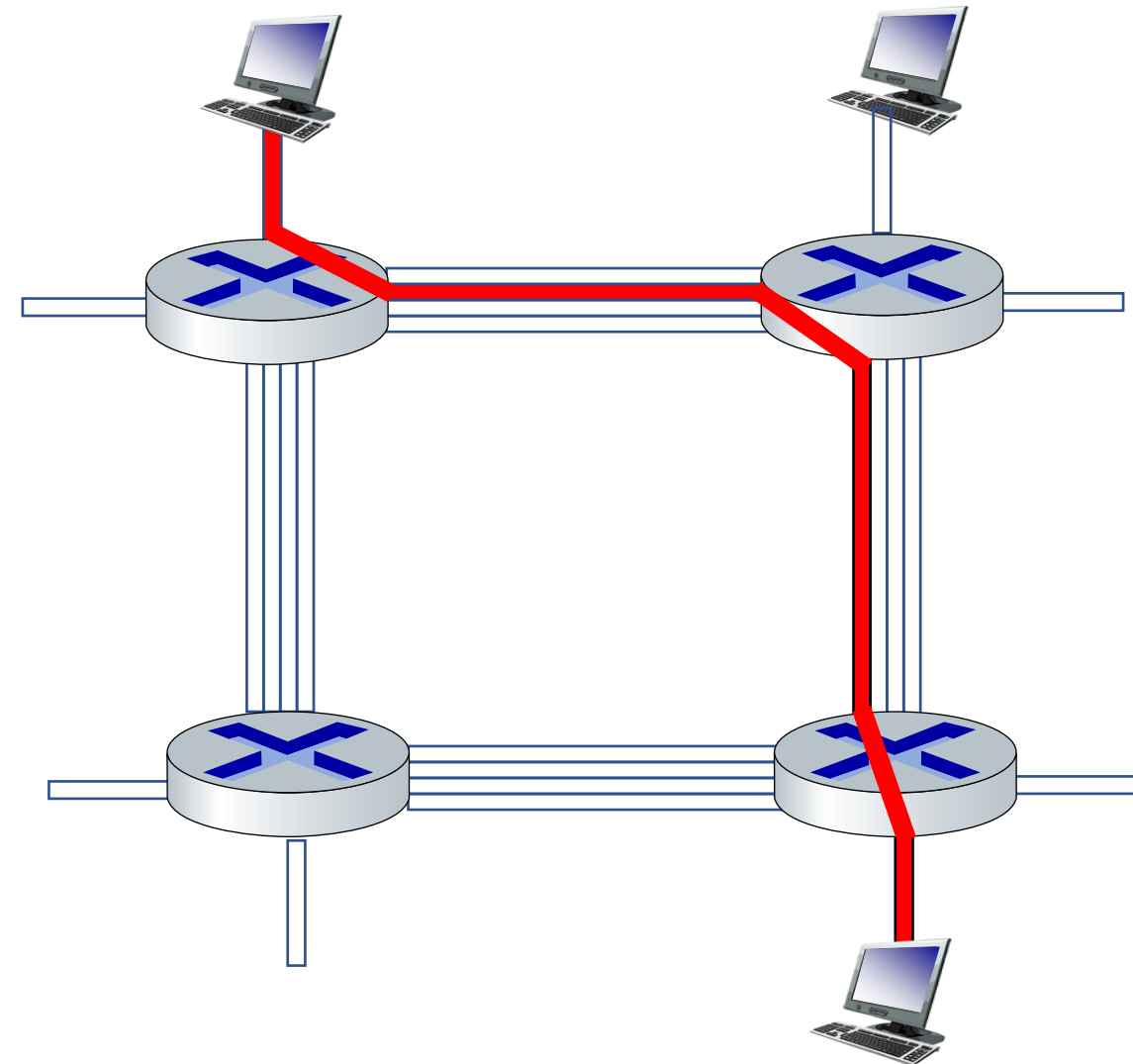
- There are two fundamental approaches to moving data through a network of links and switches: **circuit switching** and **packet switching**.
- In **circuit-switched networks**, the resources needed along a path (buffers, link transmission rate) to provide for communication between the end systems are *reserved* for the duration of the communication session between the end systems.
- In **packet-switched networks**, these resources are *not* reserved; a session's messages use the resources on demand and, as a consequence, may have to wait (that is, queue) for access to a communication link.

3.10 Alternative Core: Circuit Switching (cont.)



3.11 Alternative to Packet Switching: Circuit Switching

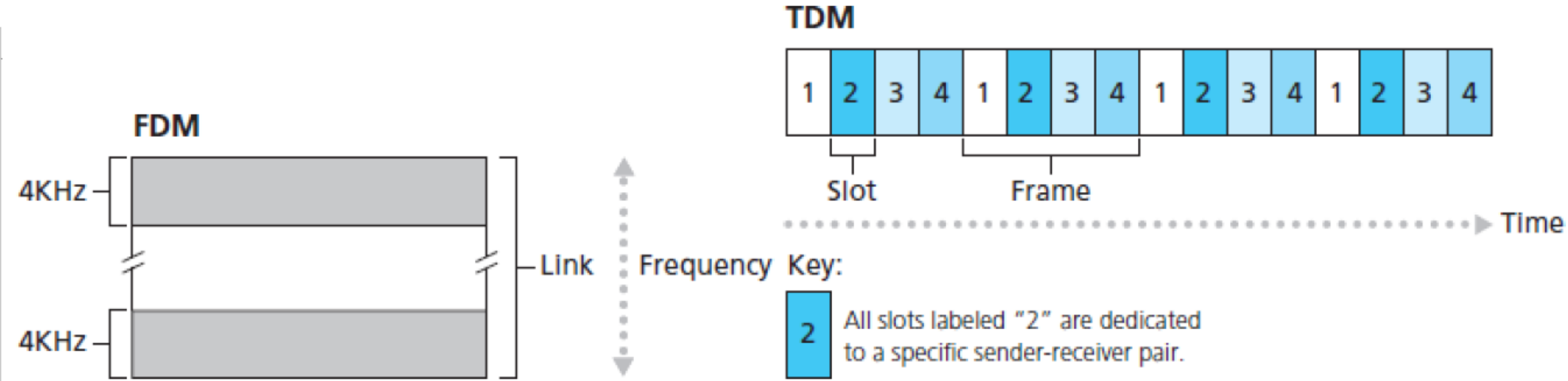
- End-end resources allocated to, reserved for “call” between source and destination.
- In diagram, each link has four circuits.
 - call gets 2nd circuit in top link and 1st circuit in right link.
- Dedicated resources: no sharing
 - circuit-like (guaranteed) performance
- Circuit segment idle if not used by call (no sharing)
- Commonly used in traditional telephone networks



3.12 Multiplexing in Circuit-Switched Networks

- A circuit in a link is implemented with either **frequency-division multiplexing (FDM)** or **time-division multiplexing (TDM)**.
- With FDM, the frequency spectrum of a link is divided up among the connections established across the link. Specifically, the link dedicates a frequency band to each connection for the duration of the connection.
- In telephone networks, this frequency band typically has a width of 4 kHz (that is, 4,000 hertz or 4,000 cycles per second). The width of the band is called, not surprisingly, the **bandwidth**. FM radio stations also use FDM to share the frequency spectrum between 88 MHz and 108 MHz, with each station being allocated a specific frequency band.
- For a **TDM** link, time is divided into frames of fixed duration, and each frame is divided into a fixed number of time slots. When the network establishes a connection across a link, the network dedicates one time slot in every frame to this connection.

3.13 FDM vs TDM

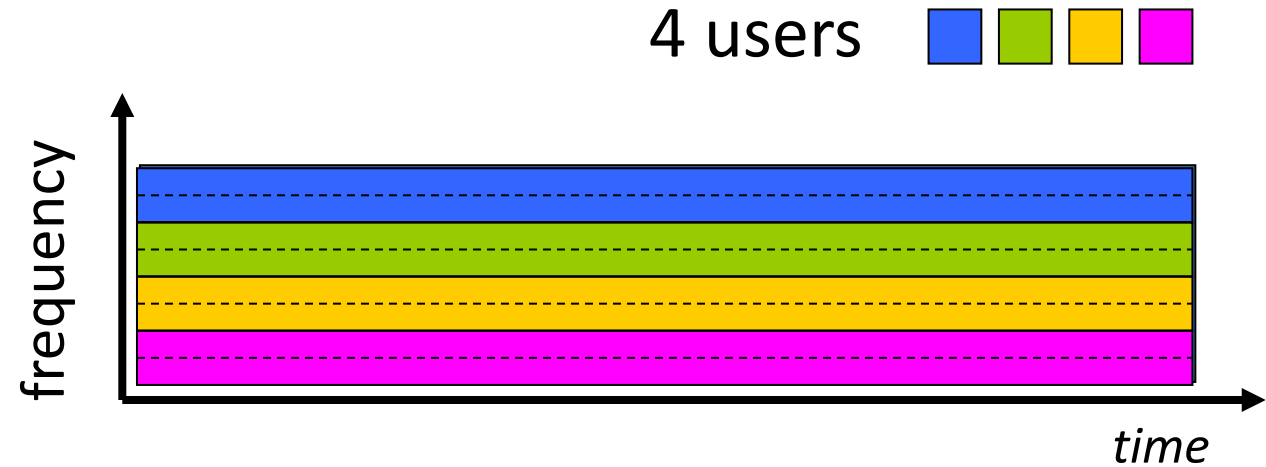


With FDM, each circuit continuously gets a fraction of the bandwidth. With TDM, each circuit gets all of the bandwidth periodically during brief intervals of time (that is, during slots)

3.14 Circuit switching: FDM and TDM

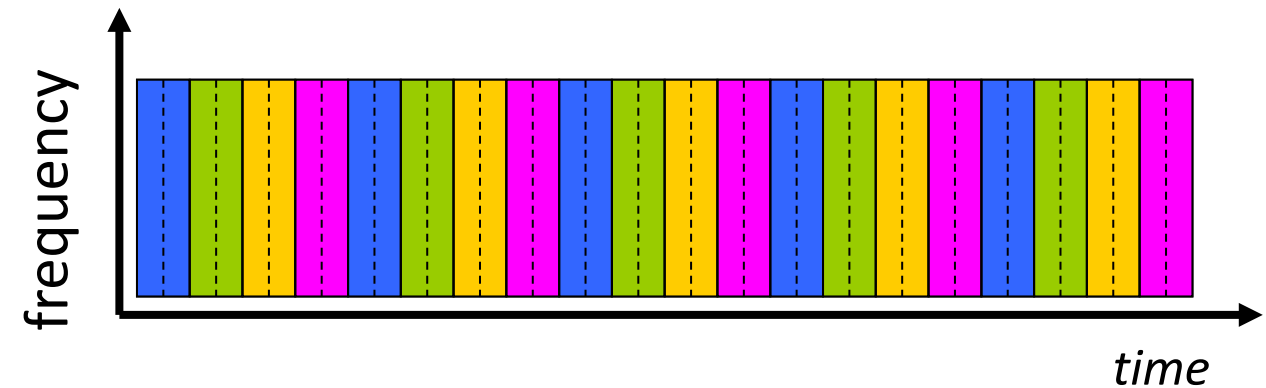
Frequency Division Multiplexing (FDM)

- optical, electromagnetic frequencies divided into (narrow) frequency bands
- each call allocated its own band, can transmit at max rate of that narrow band



Time Division Multiplexing (TDM)

- time divided into slots
- each call allocated periodic slot(s), can transmit at maximum rate of (wider) frequency band (only) during its time slot(s)



3.15 Packet Switching versus Circuit Switching

Circuit Switching

Circuit switching requires a dedicated path before sending data from source to destination.

It reserves the entire bandwidth in advance.

No store and forward transmission

Each packet follows the same route

Call setup is required

Bandwidth wastage

Packet Switching

Packet switching does not require any dedicated path to send data from source to destination.

It does not reserve bandwidth in advance.

It supports store and forward transmission

A packet can follow any route

No call setup is required

No bandwidth wastage





Week 1 - Outline

03

The Network Core

04

Next Week

4.1 Next Week

- We will focus on chapters 1.4, 1.5, 1.6, 1.7.

Your duties:

- ✓ Review this lecture note via learn.khas.edu.tr
- ✓ Find and download textbook.
- ✓ Read chapter 1 from Kurose & Ross's textbook.
- ✓ Download and Install WireShark application.



Thank You
For Your Attention!

